

Antara Chhabra

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San Diego, CA

PROFESSIONAL SUMMARY

Diligent and research-focused student with a strong foundation in research methodologies, data analysis, consistently contributing to impactful projects that drive academic advancements. Expertise in critical thinking and collaboration to solve complex problems and deliver innovative solutions. Supporting data-driven decisions, contributing effectively to dynamic team environments.

EXPERIENCE

Centre for Fundamental Research and Creative Education (CFRCE) - Research Student, Bengaluru, India (Remote) | 2024 to present

- Worked on visualization of manifolds using Python. Researched black hole's mathematical structures, thermodynamical frameworks and information paradox. Currently working on developing an agentic AI which moves towards a black hole, and its visualizations under the guidance of Dr. B. S. Ramachandra, a scientist and physicist.

Board Member and Internal Chair at Triton Quantitative Trading Club, UCSD | November 2025 to present

- Joined as a core leadership member of a student-led quantitative trading group, driving strategic decision-making, collaborative market analysis projects. Contributing to research and development of trading algorithms, supporting a data-driven approach to trading on QuantConnect platform. Contributing to design and development of a quantitative trading course. Leading a fintech project for Quantathon.

Member, CSE Society Cohorts Program, UCSD | November 2025 to present

- Invited to join the inaugural CSES Innovate Cohorts, a structured program blending research fundamentals, business development, and technical skill-building through curated workshops. Gaining exposure to entrepreneurial mindset and innovation techniques guided by industry experts, including the BlackStone LaunchPad director.

Participant, ACM AI School 2025, UCSD | October 2025 to present

- Selected among a competitive cohort for intensive training in Artificial Intelligence with a focus on Computer Vision. Engaging in hands-on group competitions and collaborative projects to deepen AI understanding and practical skill application. Developing teamwork and problem-solving skills in a cutting-edge academic environment.

Hackathons | January 2026 to present

- Hard Hack by IEEE at UCSD - Best Use of Gemini API
- SandHacks by CSES at UCSD - Overall Runner Up

EDUCATION

University of California, San Diego	2025 – 2029
• B.S. in Computer Science with Provost Honors	
• UCSD Triton Scholars Award	
Shri Ram Global School, Whitefield	2023 – 2025
• Grade 12 – 94%, Grade 11 – 97.8%	
GEAR Innovative International School, Bellandur	2009 – 2023
• Grade 10 – 97.2% in 6 subjects	

SKILLS AND CERTIFICATIONS

- Innovation, Effective leadership, Computer Programming, Data Visualization, Analytical and Critical Thinking, Complex Problem-solving
- Advanced Python, Java, NumPy, Pandas, Matplotlib, OpenCV libraries; Data aggregation, summarization and visualization of real-world dataset projects using Python and Java.

DETAILED LISTING OF HONORS SINCE HIGH SCHOOL

- International Rank-1 in International English Olympiad 2022 organized by Science Olympiad Foundations; 1st out of several thousand students from several thousand schools globally; Secured perfect score; Won Gold medal.
- International Research Olympiad Silver Medal 2024, USA (Individual category); 1000+ participants competed from 45 countries in IRO 2024 in the 3 exam stages of Prelims, Semifinals and Finals (Format: Written, Oral, Interview, Video submission); Cleared all exam stages; Was one among 15 finalists worldwide; Solely represented my country in the Finals at USA; Won Silver medal.
- International Research Olympiad Best Team Award 2024, USA; 1000+ participants from 45 countries in IRO 2024; Teams of 3 out of 15 global finalists competed in multiple rounds of defending research approaches, rapid

crossfire questions from the Jury members, presentation of innovative research ideas and findings; Best team award for demonstrating cohesiveness, collaboration and a dedication to the scientific research pursuit.

- 2x qualifier of Young Entrepreneurs Boot Camp 2022-23 (Hyderabad and Goa, India); Selected from 1000+ students from 20 states based on accomplishments and group discussion evaluations; Led the team to build prototype app on Women's safety; Pitched to jury of CEOs/Founders; Won special excellence award certificate for overall contributions; Honed leadership skills.
- Distinguished Performance ASSET certificate in math, social science, science and English exams for 2022-23, 2020-19, 2019-18, 2018-17, 2016-17; Awarded to the top 5% among several thousand students from several thousand schools participating in annual exams.
- All-Rounder Achievement Award 2024; Given to 1 in 360+ students of Grade 12 in school.
- Kalaakriti medal by Heritage Foundation of Art and Culture; Outstanding singing performance in Hindustani classical vocal in national competition 2022-23; Participants from 110 schools across 32 cities.
- National rank 29 in the Senior Category of the Hindustani Classical Vocal Music Solo competition, Vandana 2022, by Heritage Foundation of Art and Culture; Participants from 103 schools across 54 cities.
- Bestowed the title of Sangeet Bivakar for Hindustani Classical Vocal Music 2022 by Bangiya Sangeet Parishad, West Bengal, India; Graduated with Music degree; First class honors with distinction (Junior Diploma, Senior Diploma and Junior Degree); Passed 6 exams (theory and vocal practical).
- Wingword Poetry Competition Certificate 2024; Selected as one of the top 500 poets of India; Authored poem titled "The Captive" published in the Wingword Poetry anthology and online at Delhi Poetry Slam website, frequented by thousands of readers per month.
- Panelist for the URA Writers Award 2022 at GEAR School to celebrate literary creativity; Solely represented Grade 9 out of 100 students in panel discussions that included students up to Grade 12.
- Awarded the Spirit of PFS trophy for outstanding excellence in the field of academics on the Indian Independence Day 2024 by the Purva Fountain Square (PFS) Association, Bengaluru.

DETAILED LISTING OF ACTIVITIES SINCE HIGH SCHOOL

- Founding President of Research Club, SRGS School; Club aims to guide junior students towards research, innovation and creativity, research and science projects, participate in Olympiads.
- Senior School Captain (Head of Student Council) at GEAR School; Selected and elected by the school board and students based on excellence in academics and extra-curricular activities over several years; Led 1000+ students; Drafted and reviewed proposals for school authorities and events; Delivered speeches in school events and assemblies; Helped organize national and school level fests like CIQ-Mind for educators nationwide and Resurgence Economics fest for schools.
- Display of work in Annual Exhibition and School Magazines; Authored classwork, projects, poems and short stories published in school magazine and exhibited at Multiple Intelligences Annual Exhibition.
- Director/Scriptwriter/Actor; Directed 200+ school students in events and plays on topics like mental health, Indian festivals and awareness plays (Teacher's Day, Founder's Day, United Nations Day).
- Represented school in basketball, volleyball and chess in games, intra-school & inter-school matches.
- Master of Ceremonies, PFS Cultural Events, Bengaluru; Hosted large scale cultural programs and events at 2000+ residents' community; Performed dance and Hindustani Classical Vocal Music pieces during festivals and annual days.
- Basketball Team Captain at PFS; Awarded MVP (Most Valuable Player) trophy and 1st place in Annual Apartment Leagues; Strategized game plan by identifying strengths and weaknesses of opposing team; 3x Marathon runner with consistent top 5 finish.
- Organizer/Coordinator of various donation drives at PFS; Old clothes, toys, stationary for the underprivileged families; Health-camps for the security staff, maids, housekeeping (150+ people); Free food distribution seva during festivals; Building a mobile app GiveBak to streamline donations.
- Harmonium and guitar player; Composer of English and Hindi songs; Performed at community events.
- Managed and hosted various entrepreneurial stalls (baking, beverages, self-made stationary) at community events; Gained exposure to planning, budgeting, customer engagement and marketing activities.